Fiscal Year:	FY 2004	**Task Last Updated:**	FY 06/07/2004
PI Name:	Dietz, Nikolaus Ph.D.		
Project Title:	Optical Characterization of Gas Phase, Gas Phase Chemistry, and Surface Chemistry during High Pressure Vapor Phase Deposition Processes under Laminar and Turbulent Flow Conditions		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Electronic materials		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	30302-4106	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	98-HEDS-05
Start Date:	05/08/2000	**End Date:**	12/31/2004
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	2
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Bachmann, K. (North Carolina State University)		
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Performance Goal No.:			
Performance Goal Text:			
	This research task target InN as model material, which, under conditions of conventional organometallic chemical vapor deposition (OMCVD) at sub-atmospheric pressure, is limited to processing temperatures = 800 K. We focus onto deposition temperatures ? 900 K, which we consider as being essential for improving film quality and for improving the match between optimum deposition temperatures for quantum well and barrier layers of GaxIn1-xN/AlyGa1-yN multiple quantum well heterostructures. The desired range of processing temperatures mandates operating pressures of around 100 atm and above. Based on previous research and simulations of flow dynamics for selected high pressure OMCVD reactor designs we analyze and test the design of an optimized channel flow reactor for envisioned micro-gravity experiment. For process optimization as well as for real-time characterization of chemical reactions in the gas phase and at the surface, a pulsed gas handling system will be used where plugs of source vapors are separated by plugs of high pressure carrier gas. The reactor includes several optical access ports for optical methods of real-time		

Task Description:	process monitoring and to analyze the initial stages of heteroepitaxy and steady-state growth in the different pressure ranges. This research task target InN as model material, which, under conditions of conventional organometallic chemical vapor deposition (OMCVD) at sub-atmospheric pressure, is limited to processing temperatures = 800 K. We focus onto deposition temperatures _ 900 K, which we consider as being essential for improving film quality and for improving the match between optimum depo-sition temperatures for quantum well and barrier layers of GaxIn1-xN/AlyGa1-yN multiple quantum well heterostructures. The desired range of processing temperatures mandates operating pressures of around 100 atm and above. Based on previous research and simulations of flow dynamics for selected high pressure OMCVD reactor designs we analyze and test the design of an optimized channel flow reactor for envisioned micro-gravity experiment. For process optimization as well as for real-time characterization of chemical reactions in the gas phase and at the surface, a pulsed gas handling system will be used where plugs of source vapors are separated by plugs of high pressure carrier gas. The reactor includes several optical access ports for optical methods of real-time process monitoring and to analyze the initial stages of heteroepitaxy and steady-state growth in the different pressure ranges. The significance of the reactor design is that it addresses the problems of optimization of flow conditions, containment of high pressure gases and real-time monitoring of deposition/etching kinetics under conditions not accessible with any currently available commercial OMCVD reactor. High pressure growth enables the control of stoichiometry of compounds at higher temperatures that accessible at subatmospheric pressure with potential gains in perfection of the epilayers and their electrical and optical properties. It also can improve support of high density particles in fluidized bed processes for coatings, which pertain to improved fuel forms for NASA's interplanetary space exploration mission. The experiments addressing the real-time optical monitoring of indium nitride heteroepitaxy is expected to enhance the manufacturability for this important optoelectronic material in the context of advanced solid state lasers, detectors and sensors. The work will provide insights into fundamental questions of heteroepitaxy related to the chemical and structural stability of heterointerfaces and mechanisms of defect formation and control of defect propagation. The integration of real-time monitoring techniques to characterize the gas phase kinetics and thin film formation process will provide essential input parameter used in design and modeling of heterogeneous nucleation dynamics and thin film growth kinetics.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	After completing the high pressure chemical vapor deposition (HPCVD) reactor, testing / optimizing the performance of the reactor, and the evaluation of the real-time optical diagnostic techniques during the FY03, the main focus during FY04 is the application of the real-time diagnostics to characterize the flow and the gas phase decomposition kinetics of the precursors used in the growth of InN and elevated pressure. The flow characteristics has been evaluated as a function of flow rate, pressure, and temperature gradient. The process parameter and boundary conditions between laminar and turbulent flow have been documented. Detailed UV absorption spectroscopy studies on the decomposition kinetics of ammonia and Trimethylindium as function of flow, pressure and temperature have been carried out. This data are analyzed and entered in a optical data base, which will be the foundation for advanced simulations and predictions on InN nucleation and growth. Future experimental work during this research program will focus on the completion of the assessments of real-time optical diagnostics during InN nucleation and the verification of process- and growth modeling results. The HPCVD reactor simulations have been carried out using numerical solutions of nonlinear, coupled partial differential equations representing the conservation of momentum, energy and total mass as well as balances over the individual species. A three-dimensional evaluation focused on thermal distribution and gas phase reaction coupled to flow dynamics for steady state conditions, making use of the reduction in computational effort afforded by symmetry. The surface reaction mechanism for growth of InN has been described by a reduced-order model, based on pulsed chemical vapor deposition, which substantially reduces the number of reacting species. The V/III ratio has been varied and has been chosen properly to assure the incorporation of In-atoms supplied to the surface into the InN lattice. Information related to this research program are posted on the website " http://www.phy-astr.gsu.edu/ ".
Bibliography Type:	Description: (Last Updated: 03/16/2006)
Articles in Peer-reviewed Journals	Dietz, N.; Born, H.; Strassburg, M.; Woods, V. "Real-time optical monitoring of gas phase dynamics for the growth of InN at elevated pressures" Mat. Res. Soc. Symp. Proc. 798, ISBN 1-55899-736-9, pp. Y10.45.1 , Jan-2004
Articles in Peer-reviewed Journals	Dietz, N.; Born, H.; Strassburg, M.; Woods, V. "Real Time Optical Characterization of Gas Flow Dynamics in High Pressure Chemical Vapor Deposition" J. Vac. Sci. Technol. A 22 (4), pp.1-4 , Jul-2004
Articles in Peer-reviewed Journals	Cardelino B. H. , Moore C. E., Cardelino C. A., McCall S. D., Frazier D. O., Bachmann K. J. "Semiclassical calculation of reaction rate constants for homolytical dissociation reactions of interest in OMCVD" J. Physical Chemistry A, Vol. 107, pp. 3708-3718 , Jul-2003
Papers from Meeting Proceedings	Dietz N., Woods V., McCall S. D., and Bachmann K.J. "Real-time optical monitoring and simulation of gas phase kinetics in InN vapor phase epitaxy at high pressure" Proceedings of the Microgravity Conference 2002, Huntsville AL. June 25-26, NASA/CP-2003-212339, pp. 169 -181 (2003). Jan-2004
Presentation	Woods V., Born H., Strassburg M., and Dietz N. "Real-time optical monitoring of gas phase kinetics in InN Vapor Phase Epitaxy at elevated pressures" “SC4: In-Situ Monitoring of Semiconductor Growth and Processing”, AVS 50th International Symposium, Baltimore, Maryland Nov-2003
Presentation	Dietz N., Born H., Strassburg M., and Woods V. "Real-time optical monitoring of InN gas phase kinetics at elevated pressures" Dietz N., Born H., Strassburg M., and Woods V., “Real-time optical monitoring of InN gas phase kinetics at elevated pressures ”, MRS Fall Meeting in Symposium Y, Boston, MA Dec-2003

Presentation	Cardelino B. H., Moore C. E., McCall S. D., Cardelino C. A., Dietz N., Bachmann K.J. "Advanced Computational Modeling of Vapor Deposition Reactors" CAITA-2004, WestLafayette Purdue University, July 8-11, 2004, ISBN 86-7466-117-3, (2004). Jul-2004